Flinn Analytics Challenge: Commercial & Product KPIs

Project overview

This project answers five commercial and product questions using HubSpot CRM data and backend product events:

1. How many customers do we have today?
2. What is the average value of won deals, and what is the estimated new-business annual contract value (ACV)?
3. How well are users retained over time?
4. How do deals progress through the sales funnel?
5. How quickly do new organisations reach their first product value (TTFV)?

The solution uses **Snowflake** for data storage and querying, **dbt** for transformation and reusable metric models, and **Streamlit in Snowflake** for the dashboard.

> **How to read the headline numbers:** The dashboard counts a company as a customer when it has a deal that is currently marked as won and has not later been marked as lost. The source data does not contain contract start/end dates, so this is the best customer proxy available in the dataset; it does not prove that the company is still paying or has a live subscription. For deal value, €12,967.74 is simply the average amount of all won deals. The new-business average is €13,538.46. We call that an *estimated ACV* only because the new-business deals are labelled Annual License and we have assumed each amount represents one year. The data does not contain contract length to verify that assumption.

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1. Results at a glance

#	Question	Result	Source
Q1	Companies counted as customers	**26** out of 253 HubSpot companies	`MART_CUSTOMER_PORTFOLIO`
Q2	Average won deal value	**€12,967.74** across 31 won deals	`MART_REVENUE_PERFORMANCE`
Q2	Estimated new-business ACV	**€13,538.46** across 26 new-business wins, assuming deal amounts represent one year of licence value	`MART_REVENUE_PERFORMANCE`
Q3	User retention	**98.6%** Month 1; **95.9%** Month 3; **95.0%** Month 6	`MART_USER_RETENTION`
Q4	Sales win rate	**45.6%** (31 won / 68 closed); median won sales cycle **98 days**	`INT_DEAL_LIFECYCLE`
Q5	Median time to first value	**40.7 hours**; 78.4% of eligible organisations activate within 7 days; 86.1% within 30 days	`MART_TIME_TO_FIRST_VALUE`

The product event data runs through **8 February 2026**, so February 2026 is a partial month. All won deal amounts are in EUR.

Interpretation at a glance

- **Customer portfolio:** 26 companies meet the current-won-deal classification. This is a useful CRM proxy, but it does not establish that all 26 have a live contract today.
- **Deal value:** €12,967.74 is the mean value of all won deals, including new and existing business. €13,538.46 is the mean for new-business wins only and is an estimated ACV under the annual-licence assumption.
- **Retention:** the headline retention figures are user-level, monthly, weighted cohort retention. They are not customer, organisation, subscription or revenue retention.
- **Sales funnel:** the funnel uses current won status for its final stage, rather than relying solely on historical Closed Won timestamps.
- **Activation:** TTFV is measured from the recorded organisation-created event to the first subsequent search event.

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2. Data sources and scope

The challenge data contains four source datasets:

Source	Contents	Approximate volume / coverage
HubSpot companies	Company identity, name, domain, industry, country, employee count and creation date	253 companies
HubSpot contacts	Contact identity and lifecycle stage	489 contacts
HubSpot deals	Deal identity, company association, amount, currency, deal type, status, pipeline and stage timestamps	89 deals
Backend events	Organisation and user activity, event name and timestamp, search and label identifiers	39,206 events; 150 users; 37 organisations

Source-data considerations

- HubSpot and backend data are separate systems. The dashboard uses the models described below and does not assume that HubSpot company IDs and backend organisation IDs are interchangeable.
- The deal data contains 31 currently won deals, 37 lost deals and 21 open deals.

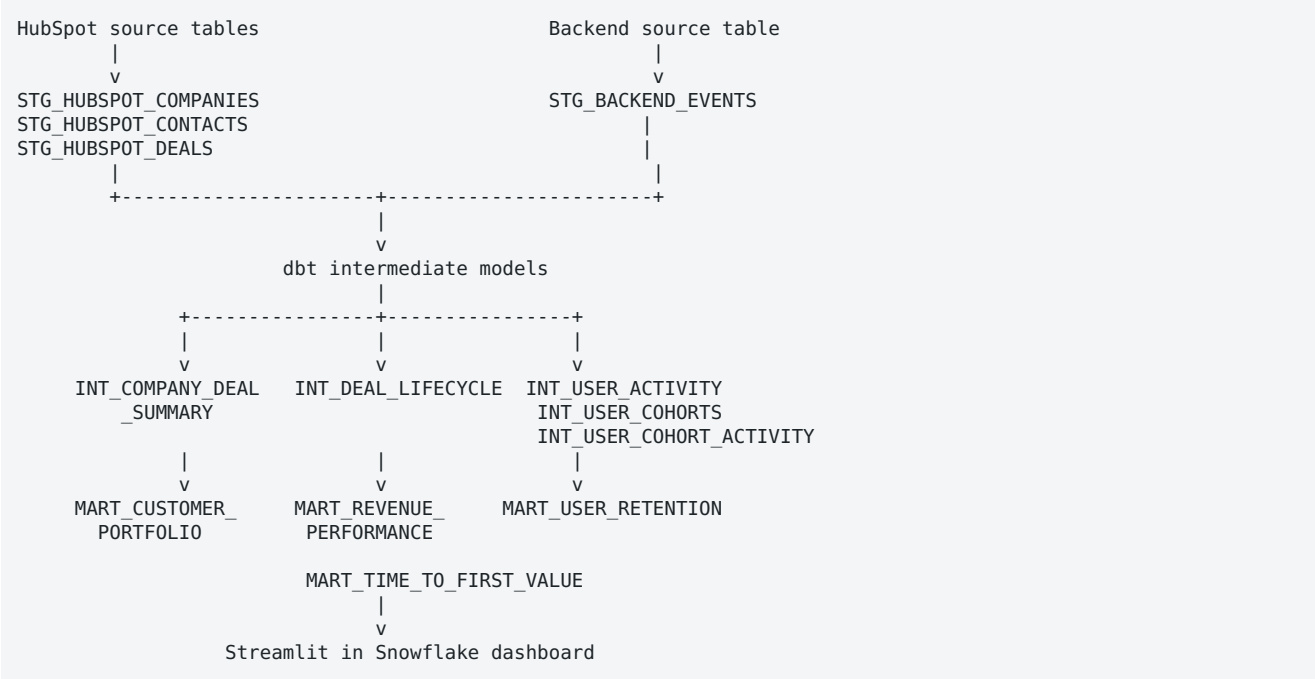
Data loading

The supplied HubSpot and backend datasets were uploaded directly into Snowflake and made available as source datasets/tables in the `FLINN_DB` database. The dbt staging models reference those Snowflake source tables. The transformation models are built in the dbt development schema `FLINN_DB.DBT_REBECCAAKOTO`.

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3. Architecture and transformation process

End-to-end flow



dbt model inventory

Layer	Model	Purpose
Staging	`STG_HUBSPOT_COMPANIES`	Standardises HubSpot company fields and types
Staging	`STG_HUBSPOT_CONTACTS`	Standardises contact fields and lifecycle stages
Staging	`STG_HUBSPOT_DEALS`	Standardises deal amounts, status, type, company IDs and stage timestamps
Staging	`STG_BACKEND_EVENTS`	Standardises event names, timestamps, identifiers and event categorisation
Intermediate	`INT_COMPANY_DEAL_SUMMARY`	Aggregates deal history and counts to one row per company
Intermediate	`INT_DEAL_LIFECYCLE`	One row per deal with status flags, stage timestamps and sales-cycle information
Intermediate	`INT_USER_ACTIVITY`	One row per user per active calendar month, using product-usage events
Intermediate	`INT_USER_COHORTS`	Assigns each user to their first product-usage month
Intermediate	`INT_USER_COHORT_ACTIVITY`	Connects users to each active month and calculates months since cohort
Mart	`MART_CUSTOMER_PORTFOLIO`	One row per HubSpot company with a customer/prospect status classification
Mart	`MART_REVENUE_PERFORMANCE`	Won-deal value and average deal value grouped by deal type and currency
Mart	`MART_USER_RETENTION`	Cohort retention by month since first product use
Mart	`MART_TIME_TO_FIRST_VALUE`	Organisation creation-to-first-search timing and activation eligibility

Why the layers are separated

- **Staging:** creates consistent, typed source models. It is intended to be close to the source and avoid embedding question-specific business logic.

- **Intermediate:** holds reusable transformations, such as deal lifecycle flags, company-level deal summaries and user-month activity.
- **Marts:** exposes business-ready data at the grain needed for each question.
- **Streamlit:** reads the marts and the deal lifecycle model to calculate display metrics, apply dashboard filters and render charts.

All models were developed in the dbt development schema `FLINN_DB.DBT_REBECCAAKOTO`. The schema name identifies the development environment; it is not a bronze/intermediate business layer.

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4. Question-by-question methodology and findings

Q1: How many customers do we have today?

Result: 26 companies, representing approximately 10.3% of the 253 companies in HubSpot.

Company status	Companies	Share	Counted as customer?
Lead	175	69.2%	No
Churned-Lost	35	13.8%	No
Active Customer	**26**	**10.3%**	**Yes**
Open New Business	15	5.9%	No
Passive Churned	2	0.8%	No
Total	**253**	**100%**	

Other portfolio indicators:

- 78 companies have at least one deal.
- 28 companies have ever had a won deal.
- 26 companies are classified as Active Customer based on current won-deal history.
- 5 of the 26 companies counted as customers have an open expansion deal.
- 15 companies have an open new-business deal.

Business definition

The portfolio mart classifies a company as:

1. **Active Customer:** has a current won deal.
2. **Passive Churned:** has a won-then-lost deal history and no current won deal.
3. **Churned-Lost:** has lost deal history but has never won.
4. **Open New Business:** has an open new-business deal and has never won.
5. **Lead:** has no deal history.

The dashboard sidebar allows the viewer to choose which CRM statuses count in the headline customer count. The default selection uses Active Customer.

Explanation: only Active Customers count as customers today: **26 of the 253 companies (10.3%)**. A company counts as a customer when HubSpot currently shows at least one of its deals as won, with no later loss overriding it. The other 227 companies are former customers, lost prospects, open prospects or leads.

Validation performed

- 253 total companies.

- 253 unique company IDs.
- 0 missing customer statuses.
- 0 companies left in Needs Review.

Q2: What is the average contract value?

ACV means the value of a contract for one year. The data gives us deal amounts, but does not tell us the contract duration. For that reason, the all-deal average is reported as an average won deal value. The new-business average is shown separately as an estimated ACV, using the assumption explained below.

Metric	Value	Interpretation
Total won deal value	€402,000	Sum of all 31 currently won deals
Average won deal value, all deal types	**€12,967.74**	€402,000 divided by 31
Average won deal value, new business	**€13,538.46**	€352,000 divided by 26
Average won deal value, existing business	**€10,000.00**	€50,000 divided by 5
Estimated new-business ACV	**€13,538.46**	Estimate, assuming new-business Annual License amounts represent one year

Deal type	Won deals	Total won value	Average	Range
New business	26	€352,000	€13,538.46	€5,000–€50,000
Existing business	5	€50,000	€10,000.00	€5,000–€15,000

Definitions

- **Average won deal value** = total amount of currently won deals divided by the number of currently won deals.
- **Estimated new-business ACV** = total amount of won new-business deals divided by the number of won new-business deals, assuming each deal amount represents one year of contract value.

Existing-business deals are reported separately because they are identified as upsells. Combining them with new-business deals could mix full contract values with incremental expansion values.

Assumption and limitation

The new-business deals are labelled Annual License. I have therefore assumed that the amount on each of those deals represents one year of licence value. On that assumption, the average is €13,538.46 per year. The dataset does not contain a contract-duration field, so this cannot be independently confirmed. If an amount covers multiple years, it would need to be divided by the number of contract years before it could be called annual contract value.

Q3: What is the retention of our users?

Results:

Months since cohort start	Weighted retention	Cohorts included	Users in eligible cohorts
Month 1	**98.65%**	17	148
Month 3	**95.92%**	16	147
Month 6	**95.04%**	13	121

Months since cohort start	Weighted retention	Cohorts included	Users in eligible cohorts
Month 12	**90.38%**	7	52

The user cohort population contains 150 users across 18 monthly cohorts.

Definitions

- **Cohort month:** the calendar month in which a user first performed a product-usage event.
- **Active user:** a distinct user with at least one qualifying product-usage event in a calendar month.
- **Month-k retention:** users from the original cohort who were active in month *k*, divided by the original cohort size.
- **Weighted retention:** total active users across eligible cohorts divided by the total cohort population of those eligible cohorts.

A cohort is included in a month-k benchmark only when it has had enough observation time to reach month *k*. This avoids treating recent cohorts as inactive simply because they have not yet reached later months.

Retention behaviour

- Early retention is high in the observed data.
- Later-month results are based on progressively fewer cohorts.
- Very small cohorts can produce large percentage movements when one user returns or becomes inactive.
- This is returning/unbounded monthly retention: a user may be inactive in one month and return in a later month. It is not consecutive-month retention.

Scope limitation: these are user-level product engagement metrics. They should not be described as paying-customer retention, organisation retention, subscription retention or revenue retention.

Data-period limitation: the backend data ends on 8 February 2026. February is incomplete, so any figure that relies on February activity may slightly understate retention and should not be compared with full calendar months.

Q4: How does the sales funnel convert?

The deal lifecycle contains 89 deals: 31 currently won, 37 lost and 21 open.

- **Win rate on closed deals:** 31 / (31 + 37) = **45.6%**.
- **Won deals as a share of all deals:** 31 / 89 = **34.8%**.
- **Median sales cycle for won deals:** 98 days.
- **Open pipeline value:** approximately €275,000.

The funnel counts deals that reached each stage:

Stage	Deals reaching it	Share of top stage	Conversion from previous stage
Pre-pitch	89	100%	—
Pitching	88	99%	99%
Product testing	82	92%	93%
Price offering	74	83%	90%
Contract negotiation	66	74%	89%
Closed won	31	35%	47%

Stage-reach rule

A deal counts as having reached a stage if it has an entry timestamp for that stage **or any later stage**. This accommodates deals that skip stages in HubSpot and keeps the cumulative funnel from widening at later stages.

Closed-won reconciliation

The raw stage timestamps contain 33 deals with a Closed Won timestamp, while the current deal status identifies 31 currently won deals. Two deals had previously entered Closed Won but were subsequently moved out of that status.

For the dashboard's final funnel stage, the current won-status flag is used. This makes the final funnel count consistent with the revenue mart's 31 currently won deals and the win-rate calculation. The two deals still count toward earlier stages they reached. This distinction is documented rather than treating historical stage entry as current won status.

Win-rate definition

Win rate = currently won deals ÷ (currently won deals + lost deals). Open deals are excluded from the denominator.

Interpretation

The largest stage-to-stage decline is from Contract negotiation to Closed won. This suggests that the closing stage merits further investigation, for example pricing, contract terms, procurement requirements and reasons for late-stage loss. The dataset alone does not establish the cause.

Q5: How quickly do organisations reach first value?

Results:

Measure	Result	Eligible organisations
Median time to first value	**40.71 hours**	37 activated organisations
Activated within 7 days	**78.38% (29/37)**	37
Activated within 30 days	**86.11% (31/36)**	36
Activated after creation at any point in the observed data	**37/37**	37

Definition

Time to first value is the elapsed time between an organisation's recorded `OrganizationCreated` event and its first `SearchExecuted` event at or after that creation timestamp.

The calculation uses the first recorded search **after** organisation creation. This is important because some organisations have search events that predate their recorded creation event, potentially due to migration or event-history timing. Those earlier events are not used to calculate TTFV.

Eligibility

- The 7-day activation rate includes organisations created at least seven days before the event-data cutoff.
- The 30-day activation rate includes organisations created at least 30 days before the cutoff.
- One organisation was not yet eligible for the full 30-day observation window, so the denominator is 36 rather than 37.

Interpretation

The median organisation reached its first recorded search in approximately 41 hours. The mean TTFV is approximately 266 hours, substantially higher than the median, indicating a long tail of organisations taking considerably longer to activate. The 7-day and 30-day rates provide more interpretable onboarding benchmarks than the overall activation rate alone.

Limitation: all 37 organisations have a post-creation search in the observed dataset. This unusually complete activation rate should be interpreted cautiously; the dataset may not include organisations that were created but never generated backend events.

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5. Data quality checks and decisions

The following issues were identified during analysis and addressed or documented.

Issue	Finding	Treatment
Historical Closed Won timestamps	33 deals had a Closed Won timestamp, but 31 were currently won	Current won status is used for the final funnel stage, revenue and win-rate metrics
Deal type naming	Values include `newbusiness` and `existing_business`	Values are kept as in the source and treated as two distinct deal types; renaming them consistently in staging is a quick follow-up
Future-dated deal	At least one deal creation timestamp was later than the data cutoff / analysis date	Flagged as a source-data quality issue; it should be investigated before time-based reporting
TTFV event order	Some first-ever searches predate the recorded organisation-created event	TTFV uses the first search at or after organisation creation
All organisations activated	All 37 observed organisations have a post-creation search	Reported, but flagged as potentially reflecting how backend records enter the dataset
Small retention cohorts	150 users across 18 cohorts	Weighted benchmarks are used; late-month results are treated cautiously
Retention can rise	Users may return after an inactive month	The metric is defined as returning/unbounded monthly retention
Partial final month	Product data ends on 8 February 2026	February 2026 is flagged as partial; figures relying on it may slightly understate retention
No contract term	No contract start/end dates or explicit term are available	ACV remains an estimate; active customer count remains a deal-history proxy
Cross-system identifiers	Backend organisation IDs do not directly equal HubSpot company IDs	No direct ID join is assumed in the dashboard KPIs

Validation performed

The following checks were used during development:

Customer portfolio

- Total company count equals 253.
- Unique company IDs equal 253.
- No missing customer status.
- No records left in the `Needs Review` fallback category.

Deal lifecycle

- 89 total deals and 89 unique deal IDs.

- 31 currently won, 37 lost and 21 open.
- No negative sales-cycle durations in the validated lifecycle output.

User activity and cohorts

- `INT_USER_ACTIVITY` contains 1,591 user-month records and 150 unique users.
- Cohort population totals 150 users.
- All 18 cohorts have a Month 0 record.
- Month 0 active users equal cohort size and Month 0 retention equals 100%.

TTFV

- 37 organisations with recorded creation events.
- 37 organisations with a post-creation search.
- The 7-day and 30-day denominators use observation-window eligibility.

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6. Dashboard

The dashboard is implemented in `dashboard/streamlit_app.py` and runs as a **Streamlit in Snowflake** application.

Dashboard tabs

Tab	Contents
Overview	Headline KPI tiles, sales funnel and user retention curve
Customers	Customer-status breakdown, company-to-customer classification funnel, industry/country breakdown and company list
Contract value	Average won deal value, estimated new-business ACV, deal-type comparison and quarterly trend
Retention	Month 1/3/6 retention, cohort selection and weighted retention curve
Sales funnel	Stage funnel, win rate, median sales cycle, open pipeline and lost-deal stage analysis
Time to first value	TTFV distribution, 7-day/30-day activation and signup-month trend

Dashboard implementation

- Connects to Snowflake using `st.connection("snowflake")`.
- Reads the dbt models from `FLINN_DB.DBT_REBECCAAKOTO`.
- Uses pandas for filtering and KPI calculations.
- Uses Altair for charts.
- Uses Plotly for the funnel when available, with an Altair fallback.
- Caches query results for 600 seconds.
- Includes a refresh button to clear cached data and rerun the app.
- Includes explanations under charts and methodology expanders for the headline metrics.

Configuration

At the top of the app, the database and model schemas are configured:

```
DATABASE = "FLINN_DB"
MART_SCHEMA = "DBT_REBECCAAKOTO"
INT_SCHEMA = "DBT_REBECCAAKOTO"
```

If the dbt target schema changes, update these values to match the schema containing the relevant models.

The dashboard reads these models:

- `MART_CUSTOMER_PORTFOLIO`
- `MART_REVENUE_PERFORMANCE`
- `MART_USER_RETENTION`
- `MART_TIME_TO_FIRST_VALUE`
- `INT_DEAL_LIFECYCLE`

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7. AI usage

AI was part of my workflow, as the brief allows. I reviewed every output against the data before keeping it.

- **Claude (Anthropic):**
  - Helped in building and iterating the Streamlit dashboard (layout, KPI tiles, funnel charts, chart explanations)
  - debugged the Snowflake connection and column-name errors
  - drafted and reviewed this README
- **What I did myself:** dbt models and metric definitions, validation queries, the final numbers and the judgement calls in the assumptions register

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8. Repository structure

```
project/
├── models/
│   ├── staging/
│   │   ├── stg_hubspot_companies.sql
│   │   ├── stg_hubspot_contacts.sql
│   │   ├── stg_hubspot_deals.sql
│   │   └── stg_backend_events.sql
│   ├── intermediate/
│   │   ├── int_company_deal_summary.sql
│   │   ├── int_deal_lifecycle.sql
│   │   ├── int_user_activity.sql
│   │   ├── int_user_cohorts.sql
│   │   └── int_user_cohort_activity.sql
│   └── marts/
│       ├── mart_customer_portfolio.sql
│       ├── mart_revenue_performance.sql
│       ├── mart_user_retention.sql
│       └── mart_time_to_first_value.sql
├── dashboard/
│   └── streamlit_app.py
├── dbt_project.yml
├── packages.yml
└── README.md
```

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9. Assumptions register

This section records the business rules and practical decisions used to turn the supplied source data into the dashboard metrics. These are not all facts contained in the source files; some are definitions chosen so that the questions can be answered consistently.

Customer portfolio assumptions

ID	Assumption / decision	How it is applied	Why it matters
A1	A customer is a company with a current won-deal relationship	A company is classified as Active Customer when it has a won deal that has not subsequently been lost	The data has no subscription table or contract end date, so this is a CRM proxy rather than proof of a live subscription
A2	A company that previously won but later lost is treated as Passive Churned	Won-then-lost deal history takes precedence over the ordinary lost-only category when there is no current won deal	Distinguishes a company that has bought before from one that never became a customer
A3	A company with only lost deals is Churned-Lost, not a former customer	Lost deals without any won history are not counted as customers	A lost prospect is not necessarily a churned customer
A4	An open new-business deal does not make a company a customer	If the company has never won and has an open new-business deal, it is Open New Business	Separates pipeline prospects from customers
A5	A company with no deal history is a Lead	No associated deal means Lead, unless another higher-priority classification applies	Provides a complete status classification for the company population
A6	The dashboard's “today” means the latest status in the supplied HubSpot extract	No historical CRM snapshot is available	The result is not a reconstruction of customer count on an earlier date
A7	Company is the grain of the customer portfolio	Each HubSpot company appears once, with its deal history aggregated before classification	Prevents companies with multiple deals from being counted more than once
A8	Dashboard customer count is configurable by status	The default counts Active Customer; the sidebar lets the viewer select other statuses	A different selection changes the headline customer count and must be stated when reporting it

Contract value and deal assumptions

ID	Assumption / decision	How it is applied	Why it matters
A9	The amount on a new-business Annual License deal represents one year of contract value	The mean of the 26 won new-business deal amounts is labelled estimated ACV (€13,538.46)	No contract-term field is available to verify that amounts are annual
A10	Overall average won deal value and ACV are different measures	€12,967.74 is the mean across all 31 won deals; it is not labelled ACV	Avoids calling a mixed deal-value average annual contract value
A11	Existing-business deals are treated as upsells and shown separately	The 5 won existing-business deals have a €10,000 average and are excluded from estimated new-business ACV	Combining upsells with new contracts could mix incremental expansion value with initial contract value
A12	Only currently won deals are included in won-value calculations	Current deal status determines whether a deal is won	Historical stage timestamps alone can include deals that are no longer won

ID	Assumption / decision	How it is applied	Why it matters
A13	Currency is handled separately	All won deals are in EUR in this dataset, so no currency conversion is applied	Amounts in different currencies should not be added without conversion
A14	Quarterly deal-value trend is grouped by the quarter in which the deal closed	Won deal amounts are averaged within close quarter	Quarters with few won deals can be volatile and should not be over-interpreted

Sales funnel assumptions

ID	Assumption / decision	How it is applied	Why it matters
A15	The funnel includes deals in the selected pipeline and filters	The dashboard's pipeline, deal type and creation-date filters determine the population in view	Funnel totals can differ when filters change
A16	Reaching a later stage implies reaching earlier stages	A deal counts at a stage if it has that stage's timestamp or a timestamp for any later stage	HubSpot users can skip stages; this rule keeps cumulative stage counts from increasing later in the funnel
A17	The final Closed Won stage uses current deal status	A deal counts as won when its current status is won, not merely because it once had a Closed Won timestamp	Two deals had a Closed Won timestamp but were no longer currently won
A18	Lost deals are analysed separately from the win funnel	Lost deals are grouped by the furthest stage reached before loss	A lost deal is not treated as another forward stage after negotiation
A19	Win rate excludes open deals	Win rate = won deals divided by won plus lost deals	Open deals have not reached an outcome; including them would measure wins as a share of all pipeline deals instead
A20	Median sales cycle is calculated for won deals	Duration is measured from deal creation to close for won deals	Open and lost deals are not included in this won-deal cycle metric

User retention assumptions

ID	Assumption / decision	How it is applied	Why it matters
A21	User ID is the retention counting unit	A user is counted once per calendar month, even if they generate multiple events	Measures people/users rather than event volume
A22	Cohort month is the first month with a qualifying product-usage event	Each user is assigned to the month of their first observed product use	The supplied data does not provide a separate, reliable user signup date for this analysis
A23	Active means at least one qualifying product-usage event in a calendar month	Authentication and account-administration events are excluded from product activity	Avoids counting a token refresh or administrative update as meaningful product engagement
A24	Retention is returning/ unbounded monthly retention	A user active in Month 1 can be counted again in Month 3 even if inactive in Month 2	The curve can rise after a decline; it is not consecutive-month retention
A25	Retention is weighted by cohort size	Active users are summed across eligible cohorts and divided by the total users in those cohorts	Larger cohorts contribute proportionately more than very small cohorts
A26	Only cohorts old enough to reach the measured month are included	Month 6 retention, for example, includes only cohorts with six months of observable history	Prevents newer cohorts from being incorrectly treated as inactive in months they have not yet reached

ID	Assumption / decision	How it is applied	Why it matters
A27	February 2026 is an incomplete month	Product data ends on 8 February 2026; it is flagged as partial wherever it affects a figure	Avoids comparing a partial period with full calendar months
A28	User retention is the chosen retention scope	Organisation-level retention and revenue/subscription retention are not included	The result answers user engagement, not commercial customer retention

Time-to-first-value assumptions

ID	Assumption / decision	How it is applied	Why it matters
A29	Organisation creation is the start of the TTFV clock	Uses the first recorded `OrganizationCreated` event for each organisation	This is the available signup proxy in backend events
A30	First value is the first search after organisation creation	Uses the earliest `SearchExecuted` timestamp greater than or equal to the creation timestamp	Some historical searches predate the recorded creation event; using them would create negative or misleading TTFV
A31	Search is the chosen first-value event	Other actions, such as exports or labels, do not define first value for this metric	A different activation event would produce a different TTFV
A32	7-day and 30-day activation rates use eligible organisations only	An organisation must have had the full 7- or 30-day observation window before the data cutoff	Prevents recent signups from being unfairly counted as not activated
A33	Overall activation means a post-creation search is present in the observed data	37 of 37 organisations in the dataset meet this condition	This does not prove every organisation ever created by the business is represented in the backend event extract

Technical and reporting assumptions

ID	Assumption / decision	How it is applied	Why it matters
A34	dbt development models are built in `FLINN_DB.DBT_REBECCAAKOTO`	The dashboard reads from that development schema	This is an environment/schema choice, not a business data layer
A35	Dashboard values are derived from dbt models and dashboard calculations	The README identifies the relevant model for each metric; some display metrics are calculated in Streamlit	The dbt model and dashboard filter context must be consistent when reconciling numbers
A36	The provided extract is the analysis population	No attempt is made to infer missing companies, users, deals or events that are not present in the supplied files	Results describe the supplied extract, not necessarily the complete live Flinn database
A37	Cross-system company-to-organisation mapping is not assumed	HubSpot company IDs and backend organisation IDs are not directly joined for the headline metrics	Avoids an unreliable join based on identifiers that belong to different systems